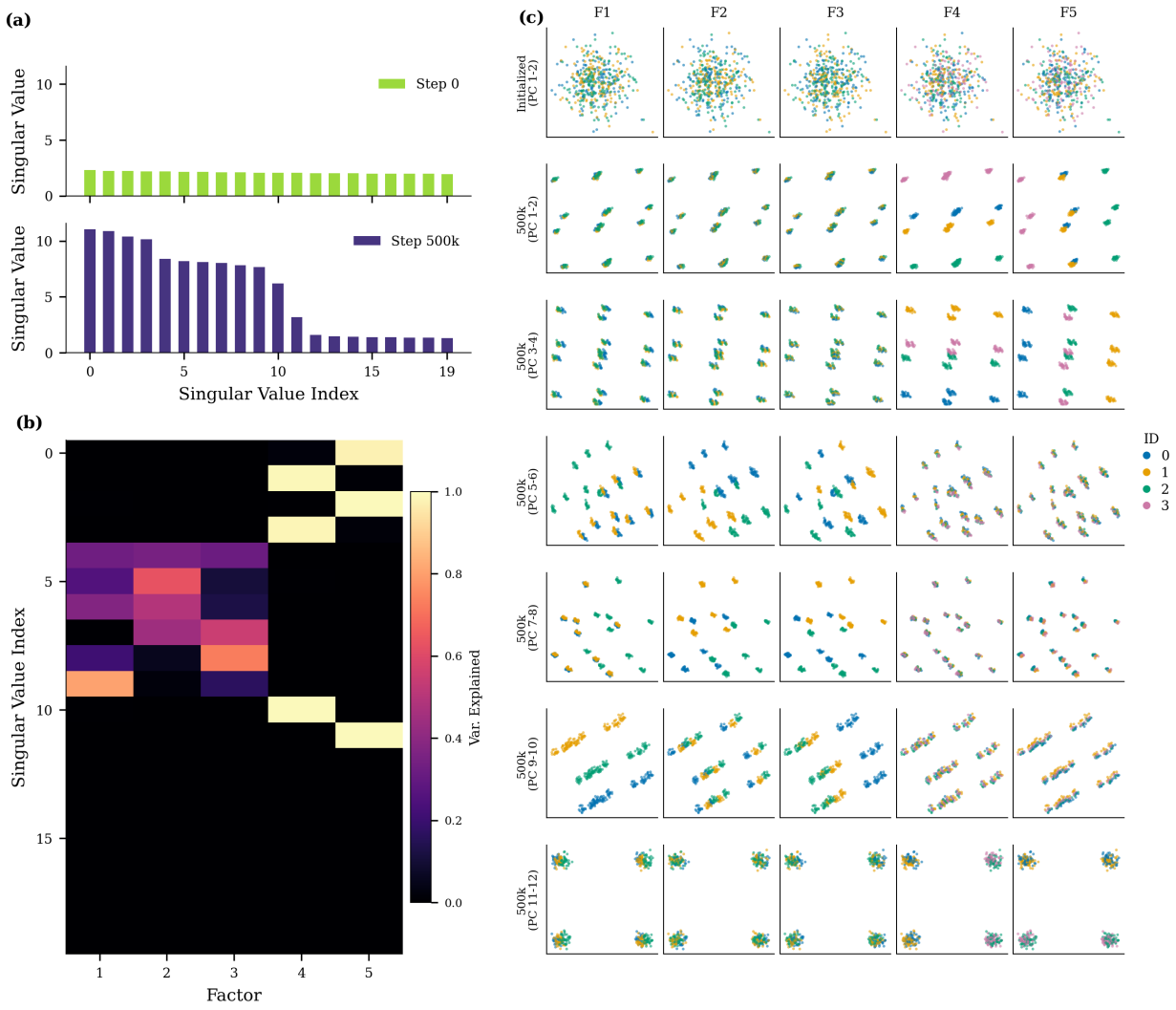

Figure 13. **The token-embedding weight matrix learns to partition the representation space according to the sub-tokens relevant to each factor.** (a) Visualizes the singular value spectrum of the embedding matrix at initialization (Step 0) versus after training (Step 500k). The trained embeddings exhibit a lower effective rank, consistent with factored structure. (b) Shows PCA projections of the token embeddings, with each point colored by sub-token identity (ID) for each factor (columns F1–F5). The top row shows randomly initialized embeddings (Step 0), which exhibit no structure. The remaining rows show trained embeddings (Step 500k) projected onto successive pairs of principal components (PC1–2 through PC11–12). We see that different PC pairs reveal different partitioning based on sub-token identities, indicating that the model learns a factored embedding with each factor encoded in a separate low-dimensional subspace. (c) Reveals the fraction of projection variance explained by each factor, for each singular vector ordered by decreasing singular value. The top 12 singular vectors show strong factor attribution (bright cells), while all subsequent rows are uniformly dark, indicating that factor-relevant structure is concentrated entirely within a low-dimensional subspace.

In the main figures we report $k^*_{0.95}$ as "Dimensions for 95%".

Technically, activations span a k^*_1-dimensional subspace. Practically, however, there are typically many fewer dimensions that are actually used appreciably. For example, only a $k^*_{0.98}$-dimensional subspace is required to explain 98% of the variance of all activations.

Tracking the evolution of k^*_p over training can be very informative about how many dimensions of the residual stream the model is using over the course of training. We find that $k^*_{0.95}$ neatly explains the discovery of low-dimensional subspaces for factored belief updating in our experiments.

We end by noting that for each point in the residual steam, we will have a different set of activations—and so we may

build an A matrix at each point (or, for that matter, any subset of points). In this work, unless noted otherwise, we look at A generated from activations associated with the final transformer block, after MLP, and before the final layer norm. For example, in our 4 layer transformer implemented using TransformerLens, we grab activations using the hook `blocks.3.hook_resid_post`.

G. Analysis: Regression

Throughout our analyses, we use linear regression to probe the relationship between transformer activations and ground-truth predictive vectors. This appendix details our methodology.

G.1. Least Squares Formulation

Given activations $A \in \mathbb{R}^{M \times d_{\text{model}}}$ from M context realizations and corresponding ground-truth predictive vectors $Y \in \mathbb{R}^{M \times d}$, we fit a linear map:

$$\hat{Y} = AW + \mathbf{1}\boldsymbol{\beta}^\top \tag{36}$$

where $W \in \mathbb{R}^{d_{\text{model}} \times d}$ and $\boldsymbol{\beta} \in \mathbb{R}^d$. The objective is:

$$\mathcal{L}(W, \boldsymbol{\beta}) = \sum_{i=1}^{M} \|Y_i - (A_i W + \boldsymbol{\beta}^\top)\|^2 \,. \tag{37}$$

To solve this, we form the augmented design matrix $X = [\mathbf{1} \mid A] \in \mathbb{R}^{M \times (1+d_{\text{model}})}$ and solve $\min_\theta \|X\theta - Y\|_{\text{F}}^2$ where $\theta = [\boldsymbol{\beta}; W]$.

G.2. SVD-Based Solution

We solve the system via singular value decomposition (SVD). Given $X = U\Sigma V^\top$, the solution is:

$$\hat{\theta} = V\Sigma^+ U^\top Y \tag{38}$$

where Σ^+ is the Moore–Penrose pseudoinverse of Σ.

For numerical stability, we apply singular value thresholding. Given a relative condition number threshold ϵ_{rcond}, we set:

$$\sigma_i^{-1} \leftarrow \begin{cases} \sigma_i^{-1} & \text{if } \sigma_i > \epsilon_{\text{rcond}} \cdot \sigma_{\max} \\ 0 & \text{otherwise} \end{cases} \tag{39}$$

where $\sigma_{\max}$ is the largest singular value. To choose a threshold, we perform 10-fold cross validation, using multiple ϵ_{rcond} values (e.g., $\{10^{-15}, 10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$) and select the threshold minimizing mean reconstruction error over the held out test sets. We then take that threshold and apply it to the full dataset to get our final fit.

G.3. Reconstruction Quality Metrics

We report reconstruction quality using root mean squared error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{Md} \sum_{i=1}^{M} \sum_{j=1}^{d} (Y_{ij} - \hat{Y}_{ij})^2} \tag{40}$$

and coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i,j} (Y_{ij} - \hat{Y}_{ij})^2}{\sum_{i,j} (Y_{ij} - \bar{Y}_j)^2} \tag{41}$$

where $\bar{Y}_j = \frac{1}{M} \sum_i Y_{ij}$ is the mean of target dimension j.

G.4. Factored Predictive Vectors

When the generative process consists of N independent factors, the ground-truth predictive vector decomposes as $Y = [Y^{(1)}|\cdots|Y^{(N)}]$ where $Y^{(n)} \in \mathbb{R}^{N \times d_n}$ is the predictive vector for factor n.

We use *joint regression*: a single regression from activations to the concatenated target Y, yielding coefficient matrix $W = [W^{(1)}|\cdots|W^{(N)}]$. This approach allows the model to use shared structure in activations for predicting multiple factors, and produces per-factor coefficient matrices $W^{(n)}$. We report both overall and per-factor RMSE.

H. Analysis: Factor Subspace Identification and Orthogonality

Our theoretical framework predicts that if transformers learn factored representations, these representations should organize into orthogonal subspaces of activation space—one for each conditionally independent latent factor of the generative process. Testing this prediction requires (1) identifying candidate factor subspaces and (2) characterizing their geometric relationships. This section first presents two procedures for factor subspace identification—vary-one analysis and regression analysis—then describes two complementary analyses that leverage these identified subspaces to quantify orthogonality. The first orthogonality analysis computes the basis-invariant pairwise overlap between factor subspaces. The second tests whether effective dimensionalities are additive: specifically, whether the sum of effective dimensionalities of factor-specific representations equals that of their union, as orthogonality would require.

H.1. Factor Subspace Identification

H.1.1. Vary-One Analysis

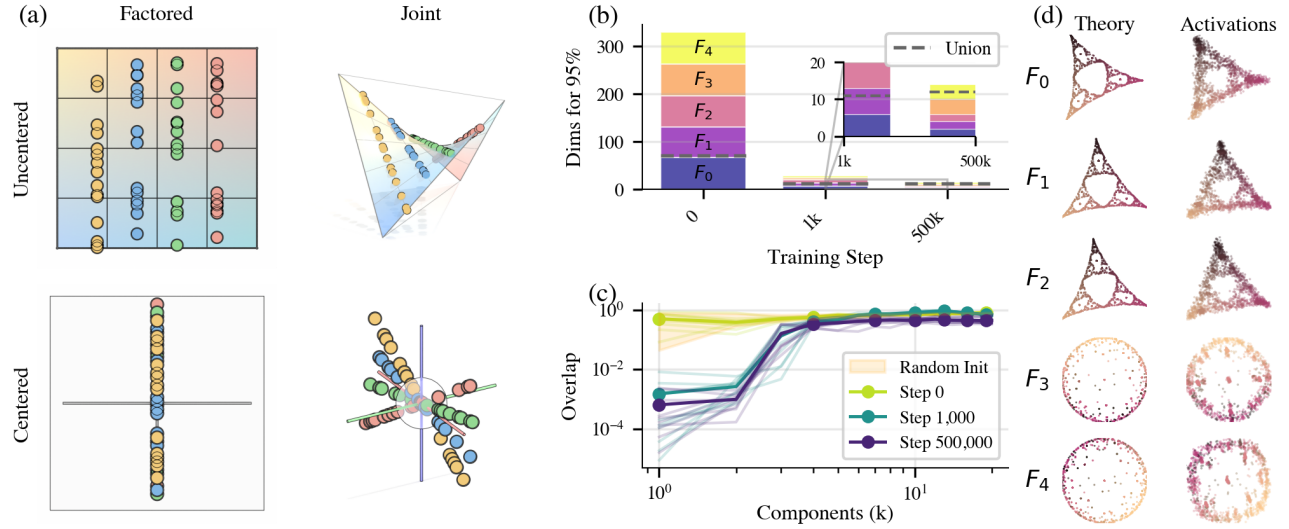

Figure 14. An expanded version of Figure 3 from the main text. a) A cartoon depiction of how we generate the vary-one dataset and how we use mean centering to isolate the variation due to each factor on its own. d) **The 2d subspaces we find through the vary-one analysis are enough to capture most of the predictive vector geometry.** Regression to the ground truth predictive vectors, using activations that come from driving the transformer on generated according to the true process and then filtering them through the first 2 PC's discovered by the vary-one analysis. Panels b) and c) are the same as in the main text.

For generative processes with N independent factors, we can generate data where only one factor's subsequences vary while the others are held fixed. This enables us to probe the dimensionality of the subspace spanned by activation variance attributable to a single factor, and to extract orthonormal bases for use in the orthogonality analyses of Section H.2.

Procedure. For each factor n, we construct datasets where only factor n's subsequences vary, keeping the subsequences of the other $N-1$ factors fixed. For each fixed configuration of the non-varied factors, we collect and mean-center the activations from an ensemble of realizations of the varied factor. We then aggregate across many fixed configurations and perform PCA. The top k principal components (chosen, e.g., to capture 95% of variance) define an orthonormal basis for a k-dimensional candidate subspace for factor n. Repeating this procedure for each factor yields N candidate subspaces with